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(SUPERSEDING)
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20 August 1958

FEDERAL SPECIFICATION

RUBBER DAM, (Dental)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rubber dam suitable for use in establishing a dry field in dental procedures.

1.2 Classification. The rubber dam shall be of the following thicknesses, as specified (see 6.1):

Thickness

Medium
Heavy
Extra heavy

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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Federal Standards:

FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-601 - Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Copies of listed federal and military standards, specifications, Commercial Item Descriptions (CIDS), handbooks and associated documents listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from the sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting activity.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D412 - Standard Test Methods for Rubber Properties in Tension.

D573 - Standard Test Method for Rubber-Deterioration in an Air Oven.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

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PAGE NOT FREE OF CUTS, TEARS, BLISTERS, AND HOLES.

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Dam shall be made from natural rubber latex having the physical requirements and conforming to the tests specified herein.

3.1.1 Color. Rubber dam shall be of a blue or green color so as to offer contrasting color to the tooth structures.

3.2 Description. Dam shall be precut into squares. Dam shall be suitable for use in mouths of dental patients to establish a dry surgical field during restorative or endodontic procedures. Dimensions of rubber dam shall be as specified in table I.

TABLE I. Dimensions.

	Length and Width (inches)	Thickness (inch)
Medium	6 ± 1/8	0.007 to 0.009
Heavy	6 ± 1/8	0.009 to 0.0115
Extra Heavy	6 ± 1/8	0.0115 to 0.0135

3.3 Physical requirements.

3.3.1 Tensile strength. Rubber shall have a tensile strength of 4000 pounds minimum per square inch, when tested as specified in 4.3.

3.3.2 Ultimate elongation. Ultimate elongation shall be a minimum of 800 percent, when tested as specified in 4.3.

3.3.3 Resistance to accelerated aging. Tensile strength and ultimate elongation shall be no less than 50 percent of the original value, when tested as specified in 4.3.

3.4 Workmanship. A slight curvature or temporary fusion of the cut edges to one another due to rotary cutting is permitted, provided separation can be accomplished easily by the operator without tearing of edges. Rubber dam shall have a smooth surface finish on both sides. Rubber dam shall be free from unpleasant odor or taste, ingrained particles, blisters, holes, tears and other imperfections which may detract from its appearance or impair its serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties of the item). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the rubber material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and III (see 4.2.1 and 4.2.2) and table IV (see 4.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table II. Unit of product for examination purposes shall be one package of rubber dams. From each package, a minimum of 3 dams shall be examined.

TABLE II. Examination Criteria.

Type defect	Inspection level	AQL (Percent defective)
Major	II	1.0
Minor	II	4.0

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4.2.2 Examination. The rubber dam shall be examined for defects including, but not limited to, those listed in table III.

TABLE III. Classification of Defects.

Categories	Defects
Major:	
101	Dams have unpleasant odor or taste.
102	Holes or tears present in dam.
103	Surface not free from ingrained particles, blisters, or other imperfections.
Minor:	
201	Precut individual rubber dams do not separate easily when taken fro package.
202	Edges not clean cut.
203	Finish not smooth.
204	Color not blue or green.

4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 inspection level S-1, AQL 4.0 (percent defective). Unit of product for test purposes shall be one precut rubber dam.

4.3 Test methods. Tests shall be conducted in accordance with FED-STD-601 or American Society for Testing and Materials (ASTM), for the applicable test method as specified in table IV.

TABLE IV. Test Methods.

Characteristic	Requirement	Test Method
Length	3.2	FED-STD-601 - 2411
Width	3.2	FED-STD-601 - 2121
Thickness	3.2	FED-STD-601 - 2011
Tensile strength	3.3.1	ASTM - D 412
Ultimate elongation	3.3.2	ASTM - D 412
Resistance to accelerated aging, at 70 ± 2 °C for 166 ± 1 hours	3.3.3	ASTM - D 573

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4.4 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit package. Thirty-six clean and dry rubber dams of one size only shall be packaged straight (not coiled or folded) in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.1.2 Intermediate package. Forty-eight units containing rubber dams of one size only shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for rubber dams shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Two intermediate packages containing rubber dams of one size only shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Waterproofing, closure, and strapping shall be as specified in the box specification.

5.2.2 Level B. Two intermediate packages containing rubber dams of one size shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing for rubber dams shall conform to MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Rubber dams packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

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5.2.4.2 Level C. Rubber dams as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, unit and intermediate packages, exterior containers, and unitized loads shall be marked as specified in FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Levels A, B, and C.

5.3.2.1.1 Unit, intermediate packages, exterior containers, and unitized loads shall be marked as specified in MIL-STD-129, except that the date of manufacture shall be shown in lieu of date packed. Marking shall include the lot or control number and the legend: "Store Between 2°-8°C (35°-46°F)."

Note: The unrefrigerated shipping time shall be not more than thirty (30) days. DD Form 1502, 1502-1, or 1502-2 as specified in MIL-STD-129 shall not be required.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Thickness required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-754-2769	RUBBER DAM, Medium, 36s
6520-00-754-2770	RUBBER DAM, Heavy, 36s
6520-01-049-0425	RUBBER DAM, Extra Heavy, 36s

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS

Preparing activity:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1835

Location S2-11-2

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